

UNIT 1 – ELECTRIC CHARGE, FORCE, AND FIELD

1.3 Charges in Uniform Fields**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. An electron enters a region of uniform field $E = 500 \text{ N/C}$ with speed $2.0 \times 10^6 \text{ m/s}$ perpendicular to the field. How far has it been deflected sideways after travelling 4.0 cm along its original direction? Ignore gravity.

$$\begin{aligned} \text{Time in the field: } t &= \frac{0.040}{2.0 \times 10^6} = 2.0 \times 10^{-8} \text{ s.} \quad \text{Acceleration: } a = \frac{eE}{m_e} = 8.8 \times 10^{13} \text{ m/s}^2. \\ \text{Deflection: } y &= \frac{1}{2}at^2 = \frac{1}{2}(8.8 \times 10^{13})(2.0 \times 10^{-8})^2 = 1.8 \times 10^{-2} \text{ m} = 1.8 \text{ cm.} \end{aligned}$$

2. An electric dipole with dipole moment $p = 6.0 \times 10^{-30} \text{ C}\cdot\text{m}$ sits in a uniform field $E = 1.0 \times 10^5 \text{ N/C}$, making a 30° angle with the field. Find the torque on the dipole and its potential energy.

$$\tau = pE \sin \theta = (6.0 \times 10^{-30})(1.0 \times 10^5) \sin 30^\circ = 3.0 \times 10^{-25} \text{ N}\cdot\text{m}. \quad U = -pE \cos \theta = -5.2 \times 10^{-25} \text{ J}.$$